Ponnu Soman

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PROFILE

Data Scientist and Analyst with an MSc in Data Science and proven experience building end-to-end machine learning pipelines, automated data engineering systems, and interactive business intelligence dashboards. Delivered production-grade predictive models and statistical experiments across energy, retail, and public sector domains — translating complex datasets into decisions stakeholders act on. Proficient in Python, R, SQL, Power BI, and Tableau.

EDUCATION

- **Master's in data science**, Seattle University 2023 – 2025 | Seattle, USA
  Coursework: Statistical Machine Learning, Hypothesis Testing, Regression Analysis, Data Visualization, SQL, Data Management, NLP, Web Analytics. GPA : 3.57
- **Integrated MSc Physics and Mathematics**, Amrita Vishwa Vidyapeetham 2017 – 2022 | Kerala, India
  Mathematical Modeling: Applied advanced mathematics and differential equations to model physical systems and simulate complex phenomena.
  Relevance: Developed strong analytical critical thinking skills rooted in physics and quantitative analysis. GPA : 3.60

TECHNICAL SKILLS

- **Languages & Databases:** Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), R, SQL (PostgreSQL, MySQL), DAX
- **BI & Visualization:** Power BI, Tableau, Streamlit, Plotly, Looker Studio, Excel (Advanced)
- **Machine Learning:** Random Forest, XGBoost, SVM, Logistic Regression, Decision Trees, K-Means Clustering, Hierarchical Clustering, PCA, Prophet, TensorFlow, Keras
- **Data Engineering:** dbt Core, Prefect, Snowflake, Azure PostgreSQL, SQLAlchemy, Isolation Forest, Git, GitHub
- **Core Competencies:** A/B Testing, Anomaly Detection, Feature Engineering, Hypothesis Testing, Dashboard Development, Data Pipeline Design, Stakeholder Communication, Data Quality and Governance

EXPERIENCE

Student Data Scientist (Capstone project) Jan 2025 – June 2025 | Seattle, WA
Snohomish County Office of Energy and Sustainability

- Led Power BI dashboard development for Snohomish County's Low-Income and Disadvantaged Communities (LIDAC) equity analysis system; connected directly to Azure PostgreSQL with custom DAX formulas, dynamic ZIP code map filtering, and real-time response density visualizations — replacing fragmented manual PDF reporting workflows across multiple county departments.
- Designed a four-component interactive dashboard (survey selection, geographic filtering, question-level response analysis, theme visualization) that automated both quantitative and qualitative community survey reporting, eliminated reliance on external consultants, and enabled county decision-makers to access timely equity insights for resource allocation and policy development.

PROJECTS

1. **E-Commerce Data Pipeline with ML Forecasting | Python, Snowflake, dbt Core, Prefect, Prophet, Isolation Forest, Streamlit, Groq API**
   - Architected a fully automated end-to-end e-commerce data pipeline — synthetic order ingestion into Snowflake via idempotent MERGE statements, dbt transformation through Staging and Mart layers (6 tables, 36 data quality tests all passing), and Prefect daily orchestration chaining ingestion, dbt build, and anomaly detection in sequence across all five pipeline phases.
   - Deployed two Prophet forecasting models (30-day revenue and 7-day per-product demand across 20 products), Isolation Forest anomaly detection with severity-coded logging, and a four-page Streamlit dashboard with a Groq-powered AI assistant that generates and executes live SQL against Snowflake — delivering a production-grade system with forecast accuracy tracked over time.
2. **New Hampshire Hourly Electricity Demand Forecasting | Python, Scikit-learn, Random Forest, Pandas**
   - Built and cleaned a 26,304-record hourly dataset integrating three years of New Hampshire power consumption data (2020–2022) with meteorological variables; engineered non-linear interaction features including dry bulb squared, dew point squared, and their interaction term to capture complex temperature-demand relationships missed by linear models.
   - Systematically compared four predictive models from linear regression (R^2=0.602) to Random Forest ensemble; final model with temporal feature engineering achieved R^2=0.894, MAE=45.19, RMSE=59.84 — a 45% reduction in prediction error over baseline — validated with 5-fold cross-validation and recommended for quarterly operational retraining.
3. **Customer Churn Prediction | Python, Scikit-learn, Logistic Regression, Pandas**
   - Analyzed a telecommunications dataset of 7,043 customers (26.6% churn rate, 21 features) to identify at-risk subscribers before cancellation; performed exploratory data analysis revealing bimodal tenure distribution and differential churn propensity across contract type, service category, and financial burden variables.
   - Implemented logistic regression with balanced class weighting to address the 73/27 class imbalance; achieved 80.65% accuracy and 85.2% ROC-AUC; identified contract duration, service type, and customer tenure as dominant churn drivers — producing risk-stratified cohorts that enable targeted retention campaigns before revenue loss occurs.
4. **A/B Testing: Website Background Color Impact | Python, SciPy, Chi-Square Test, Mann-Whitney U, Pandas**
   - Designed and executed a randomized controlled trial across 5,000 participants split between control and treatment conditions; selected statistical tests empirically using Shapiro-Wilk normality assessment and Levene variance testing to determine appropriate parametric versus non-parametric inference for each outcome variable.
   - Applied chi-square testing on the primary conversion outcome; treatment group achieved 14.07% conversion vs 5.40% control — a statistically significant 61% relative lift ($p<0.001$) — with heterogeneous subgroup analysis confirming consistent effect across all device types and geographic segments, delivering causal evidence sufficient to recommend full deployment.